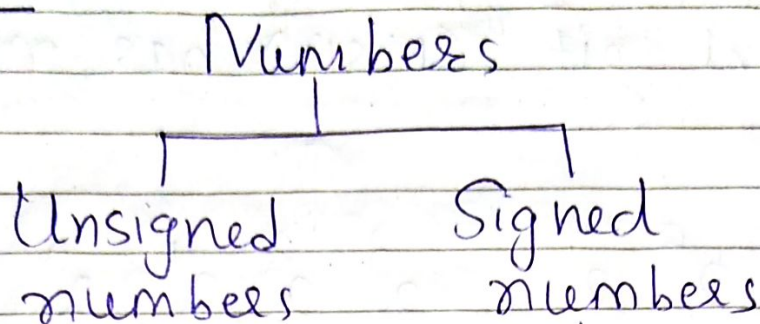


Detect if two integers have
Logic: Opposite Signs

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↓
Includes both +ve
and -ve numbers

- Signed numbers can be represented in many ways
- Usually, 2's Complement method is used for that.
- E.g. 8 bit Unsigned numbers

$$\begin{array}{cccccc} 128 & 64 & 32 & 16 & 8 & 4 & 2 & 1 \\ (0 & 0 & 0 & 0 & 0 & 0 & 0 & 0)_2 \rightarrow 0 \end{array}$$

$$(1 & 1 & 1 & 1 & 1 & 1 & 1 & 1)_2 \rightarrow 255$$

For Unsigned 8-bit number,
we have range from
0 to 255.

→ For 8 bit Signed numbers, Most Significant bit (MSB) has negative weight.

$$-2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$$

i.e.

Weight: $-128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1$

$$(0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)_2 \rightarrow 0$$

$$(1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1)_2 \rightarrow (-1)$$

$$= -128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$$

$$= -128 + 127$$

$$= -1$$

Ex: $1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0$
 $\searrow$
 -128

Ex: $0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1$
 $\searrow$
 127

Ex: $0 \quad 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $64 \quad 16 \quad 8 \quad 2 \quad 1$

$$64 + 16 + 8 + 2 + 1 = 91$$

Indicates

Note: When MSB (Sign of Number) is 0 then Number is +ve

→ When MSB is 1, Number is -ve

★ If two positive numbers are XORed or two negative numbers are XORed then Result has 0 in its MSB.

Ex:

		-128	64	32	16	8	4	2	1
64		0	1	0	0	0	0	0	0
127		0	1	1	1	1	1	1	1

Bitwise
XOR

0

0 1 1 1 1 1 1 1

Truth table of XOR		a	b	XOR(a,b)
		0	0	0
		0	1	1
		1	0	1
		1	1	0

Ex:

-1

-5

		-128	64	32	16	8	4	2	1
-1		1	1	1	1	1	1	1	1
-5		1	1	1	1	1	0	1	1
		0	0	0	0	0	1	0	0

Note: When one +ve and one -ve number is XORed then result is having 1 in its MSB.

E.g.

	-128	64	32	16	8	4	2	1
+35	0	0	1	0	0	0	1	1
-100	1	0	0	1	1	1	0	0
XOR	1	0	1	1	1	1	1	1

→ (1001 1100)₂

$$\begin{aligned} -128 + 16 + 8 + 4 &= -128 + 28 \\ &= -100 \end{aligned}$$

→ Now, this fact can be used to check if two numbers are of same sign or not.

(1) If XOR of a and b is positive number then a and b are of same sign.

(2) If XOR of a and b is negative number then a and b are of opposite sign.

→ If you are using 32 bits to represent number (signed numbers) then same idea will work.

★ Reverse Number

(fixed for 3 digits)

$$n = 125$$



$$125 \% 10 = 5 \text{ (Last digit of 125)}$$

$$125 / 10 = 12 \text{ (Remaining number)}$$



$$12 \% 10 = 2$$

$$12 / 10 = 1$$



$$5 * 100 + 2 * 10 + 1$$

$$= 500 + 20 + 1$$

$$= 521$$

→ Reverse of 125

* int num = 5

0000 0000 0000 0000 0000 0000 0101 : 5

↓ Shift Left by 1 bit

0000 0000 0000 0000 0000 0000 1010 : 10

↓ Shift Left by 1 bit

0000 0000 0000 0000 0000 0000 0100 : 20

↓ Shift Left by 1 bit

0000 0000 0000 0000 0000 0000 0010 : 40

↓ Shift Left by 1 bit

0000 0000 0000 0000 0000 0000 0100 : 80

i.e. ↓ Shift Left by 4 bits
is equivalent to multiply
by 2^4

$$5 * 2^4 = 80$$

* Similarly Shift Right by 1 bit
is equivalent to divide by 2

$$* \quad 15 \wedge 12$$

$$\begin{array}{r} 1111 \\ 1100 \\ \hline 0011 \rightarrow 3 \end{array}$$

$$* \quad 15 / 12$$

$$\begin{array}{r} 1111 \\ 1100 \\ \hline 1111 \rightarrow 15 \end{array}$$

$$* \quad 15 \& 12$$

$$\begin{array}{r} 1111 \\ 1100 \\ \hline 1100 \rightarrow 12 \end{array}$$

$$* \quad 0x0f \rightarrow (1111)_2$$

$$0x02 \rightarrow (0010)_2$$

$$\sim 0x02 \rightarrow (1101)_2$$

$$0x0f \& \sim 0x02$$

$$\& \begin{array}{r} 1111 \\ 1101 \\ \hline 1101 \rightarrow 13 \end{array}$$

* num & (1 << 3)

→ 1 << 3

↳ shift 1 by 3 bits to left

↳ This is same as

$$1 * 2^3$$

$$= 8$$

→ num & 8

Say num = 12

8 → (1000)₂

$$\begin{array}{r} \text{3rd bit} \swarrow \\ 1100 \quad \swarrow \text{0th bit} \\ \& \quad 1000 \\ \hline 1000 \\ \downarrow \\ \text{3rd bit.} \end{array}$$